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GATE ECE · CONTROL SYSTEMS · CHAPTER-WISE PYQS

# Signal Flow Graph and Block Diagram

GATE ECE Control Systems · Previous Year Questions with Answer Key

| 23 | 19 | 1987 – 2026 | English |
|---|---|---|---|
| QUESTIONS | PAPERS COVERED | YEARS | LANGUAGE |

INSIDE THIS PDF

| | |
|---|---|
| Marks 1 | 9 questions |
| Marks 2 | 11 questions |
| Marks 5 | 3 questions |
| Answer key | at the end |

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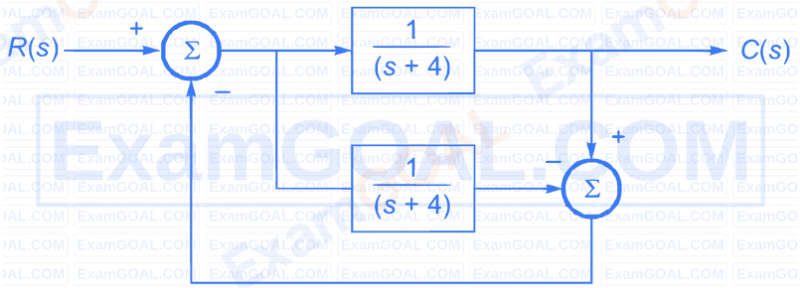
1. **Solve** — Print it, or work through it on screen.
2. **Check** — The answer key is on the last pages.
3. **Understand** — Scan the QR code beside any question for its full solution.

## Marks 1 9 questions

**Q1.** GATE ECE 2026

A control system is shown in the Figure.

Which option represents the correct transfer function of the system?




(A) $\frac{1}{(s+4)^2}$

(B) $\frac{1}{(s+4)}$

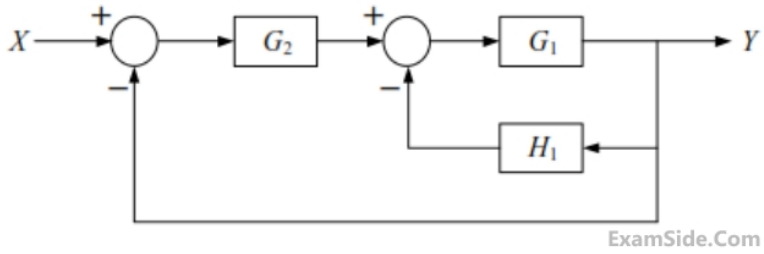
(C) $\frac{2}{(s+4)}$

(D) $\frac{1}{(s^2+8s+17)}$

Open solution

**Q2.** GATE ECE 2016 Set 3

The block diagram of a feedback control system is shown in the figure. The overall closed-loop gain G of the system is




(A) $G = \frac{G_1G_2}{1+G_1H_1}$

(B) $G = \frac{G_1G_2}{1 + G_1G_2 + G_1H_1}$

(C) $G = \frac{G_1G_2}{1 + G_1G_2H_1}$

(D) $G = \frac{G_1G_2}{1 + G_1G_2 + G_1G_2H_1}$

Open solution

**Q3.** GATE ECE 2015 Set 2

By performing cascading and/or summing/differencing operations using transfer function blocks $G_1(s)$ and $G_2(s)$, one CANNOT realize a transfer function of the form

(A) $G_1(s)G_2(s)$

(B) $\frac{G_1(s)}{G_2(s)}$

(C) $G_1(s)\left(\frac{1}{G_1 s} + G_2(s)\right)$

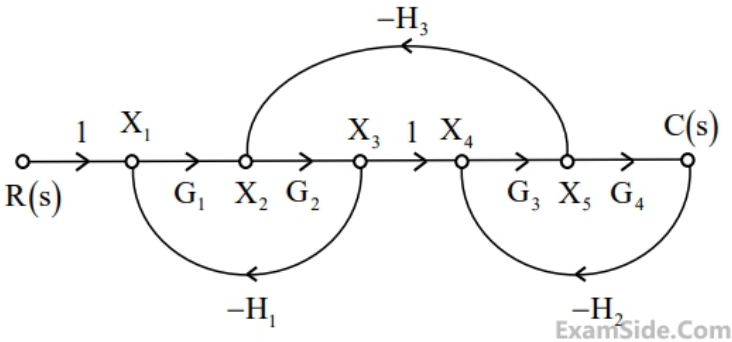
(D) $G_1(s)\left(\frac{1}{G_1 s} - G_2(s)\right)$

Open solution

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**Q4.** GATE ECE 2015 Set 2

For the signal flow graph shown in the figure, the value of $\frac{C(s)}{R(s)}$ is




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Open solution

(A) $$\frac{G_1G_2G_3G_4}{1-G_1G_2H_1-G_3G_4H_2-G_2G_3H_3+G_1G_2G_3G_4H_1H_2}$$

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(B) $$\frac{G_1G_2G_3G_4}{1+G_1G_2H_1+G_3G_4H_2+G_2G_3H_3+G_1G_2G_3G_4H_1H_2}$$

ExamSide.Com

(C) $$\frac{1}{1+G_1G_2H_1+G_3G_4H_2+G_2G_3H_3+G_1G_2G_3G_4H_1H_2}$$

ExamSide.Com

(D) $$\frac{1}{1-G_1G_2H_1-G_3G_4H_2-G_2G_3H_3+G_1G_2G_3G_4H_1H_2}$$

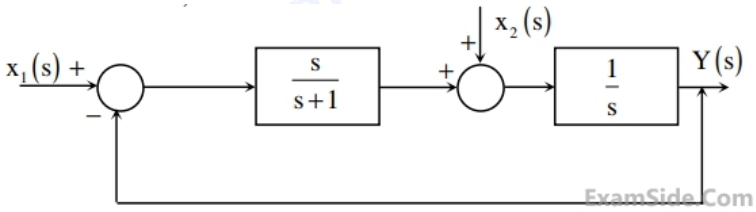
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**Q5.** GATE ECE 2014 Set 2

Open solution

For the following system,




When $X_1(s) = 0$ , the transfer function $\frac{Y(s)}{X_2(s)}$ is

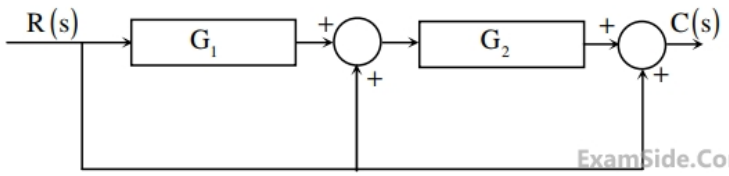
(A) $\frac{s+1}{s^2}$ (B) $\frac{1}{s+1}$

(C) $\frac{s+2}{s(s+1)}$ (D) $\frac{s+1}{s(s+2)}$

**Q6.** GATE ECE 2014 Set 3

Open solution

Consider the following block diagram in the figure.



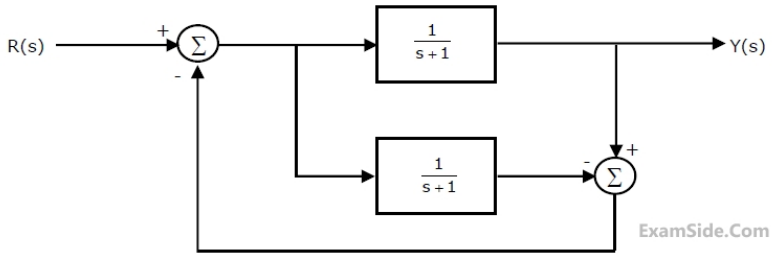

The transfer function $\frac{C(s)}{R(s)}$ is

(A) $\frac{G_1G_2}{1+G_1G_2}$ (B) $G_1G_2 + G_1 + 1$ (C) $G_1G_2 + G_2 + 1$ (D) $\frac{G_1}{1+G_1G_2}$

**Q7.** GATE ECE 2010

Open solution

The transfer function Y(s)/R(s) of the system shown is



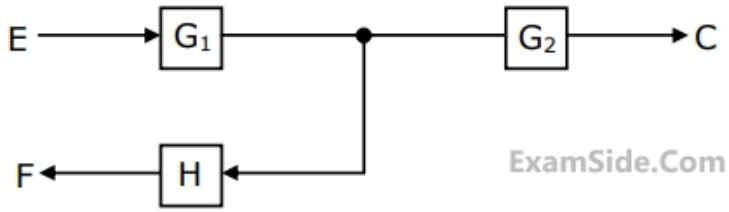

(A) $0$ (B) $\frac{1}{s+1}$ (C) $\frac{1}{s+2}$ (D) $\frac{2}{s+3}$

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**Q8.** GATE ECE 2001

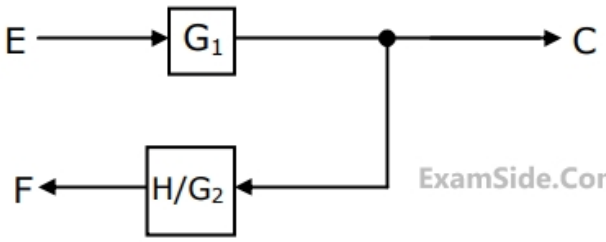
Open solution

The equivalent of the block diagram in Figure is given as



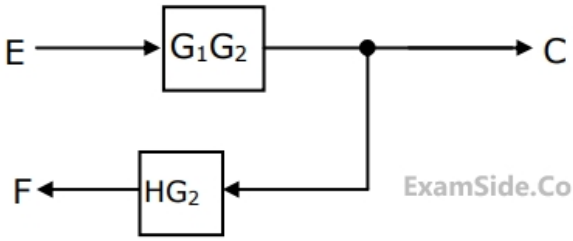

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(A)



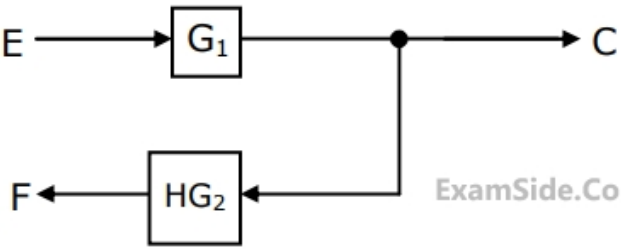

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(B)



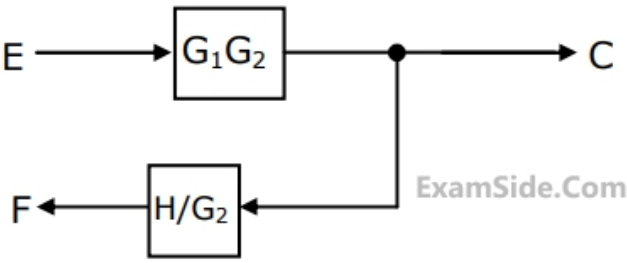

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(C)




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(D)




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**Q9.** GATE ECE 1995

Signal flow graph is used to find

(A) stability of the system

(B) controllability of the system

(C) transfer function of the system

(D) poles of the system

Open solution

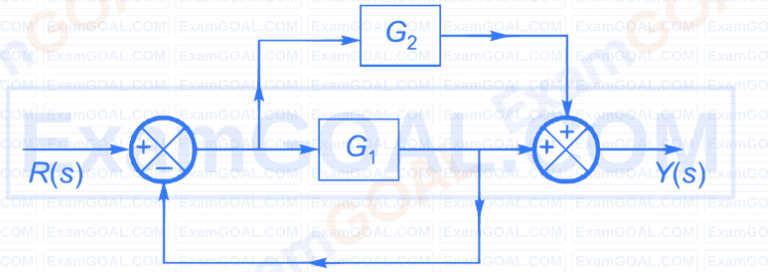
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## Marks 2 11 questions

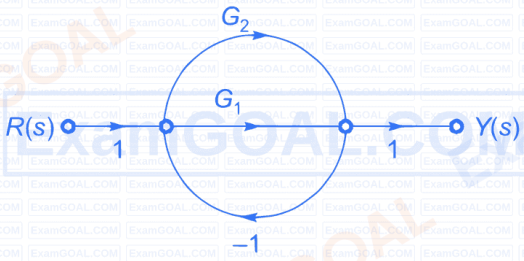
**Q10.** GATE ECE 2025

Open solution

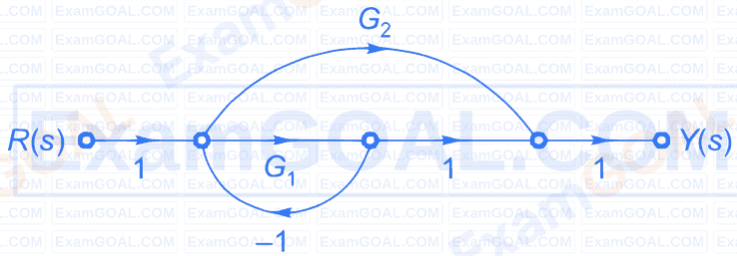
Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) this system? Choose the correct option(s).



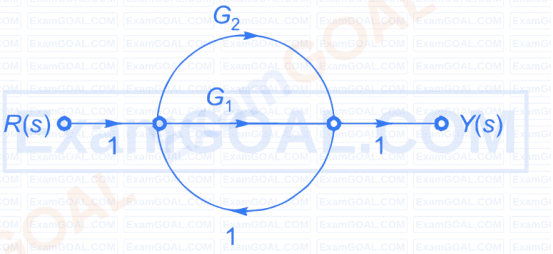

(A)



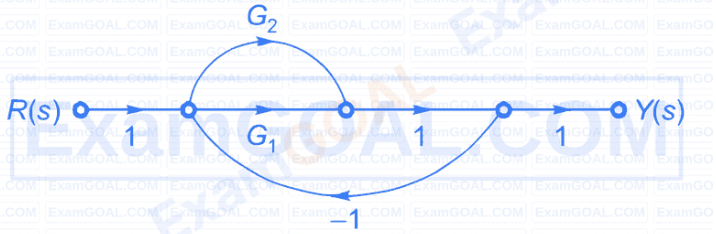

(B)




(C)




(D)



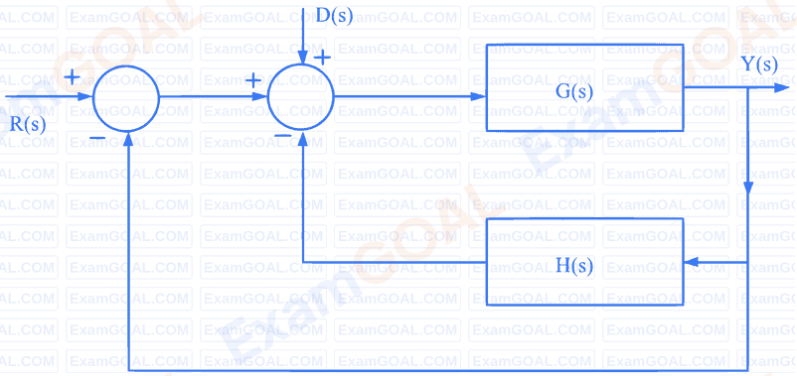

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**Q11.** GATE ECE 2023

Open solution

In the following block diagram, R(s) and D(s) are two inputs. The output Y(s) is expressed as Y(s) = $G_1(s)R(s) + G_2(s)D(s)$.

$G_1(s)$ and $G_2(s)$ are given by




(A) $G_1(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$ and $G_2(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$

(B) $G_1(s) = \frac{G(s)}{1+G(s)+H(s)}$ and $G_2(s) = \frac{G(s)}{1+G(s)+H(s)}$

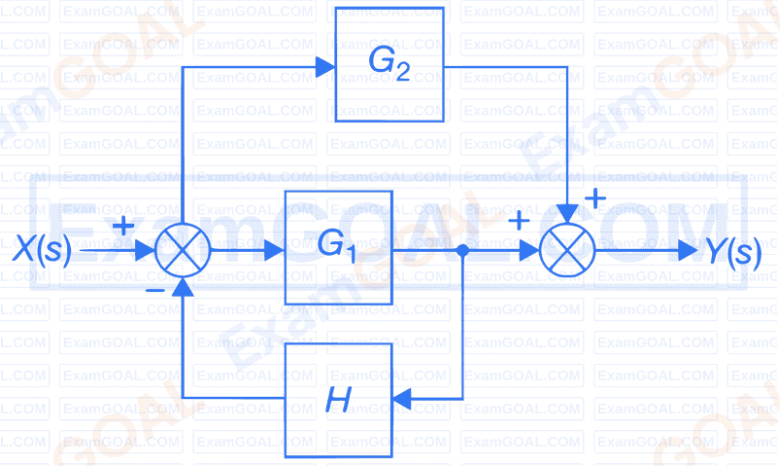
(C) $G_1(s) = \frac{G(s)}{1+G(s)+H(s)}$ and $G_2(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$

(D) $G_1(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$ and $G_2(s) = \frac{G(s)}{1+G(s)+H(s)}$

**Q12.** GATE ECE 2021

Open solution

The block diagram of a feedback control system is shown in the figure.




The transfer function $\frac{Y(s)}{X(s)}$ of the system is

(A) $\frac{G_1+G_2}{1+G_1H+G_2H}$

(B) $\frac{G_1+G_2}{1+G_1H}$

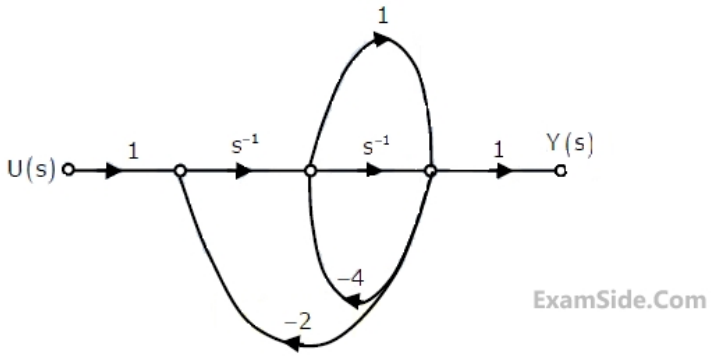
(C) $\frac{G_1+G_2+G_1G_2H}{1+G_1H}$

(D) $\frac{G_1+G_2+G_1G_2H}{1+G_1H+G_2H}$

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**Q13.** GATE ECE 2013

The signal flow graph for a system is given below. The transfer function $\frac{Y(s)}{U(s)}$ for this system is

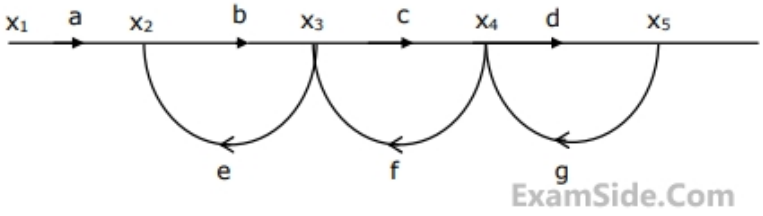



Open solution

(A) $\frac{s+1}{5s^2+6s+2}$ (B) $\frac{s+1}{s^2+6s+2}$ (C) $\frac{s+1}{s^2+4s+2}$ (D) $\frac{1}{5s^2+6s+2}$

**Q14.** GATE ECE 2004

Consider the signal flow graph shown in Figure. The gain $\frac{x_5}{x_1}$ is




Open solution

(A) $\frac{1-(be\ +\ cf\ +\ dg)}{abcd}$

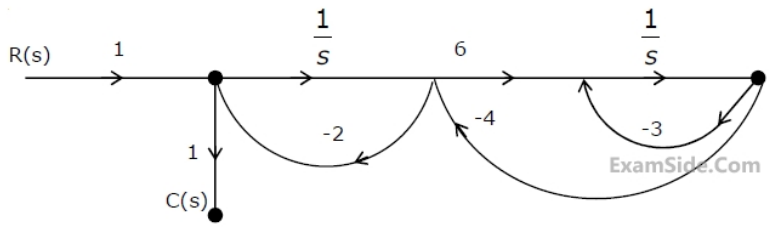
(B) $\frac{bedg}{1-(be\ +\ cf\ +\ dg)}$

(C) $\frac{abcd}{1-(be\ +\ cf\ +\ dg)\ +\ bedg}$

(D) $\frac{1-(be\ +\ cf\ +\ dg)\ +\ bedg}{abcd}$

**Q15.** GATE ECE 2003

The signal flow graph of a system is shown in figure. The transfer function $\frac{C(s)}{R(s)}$ of the system is




Open solution

(A) $\frac{6}{s^2+29s+6}$ (B) $\frac{6s}{s^2+29s+6}$ (C) $\frac{s(s+2)}{s^2+29s+6}$ (D) $\frac{s(s+27)}{s^2+29s+6}$

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**Q16.** GATE ECE 2001

Open solution

An electrical system and its signal-flow graph representations are shown in Figure (a) and (b) respectively. The values of $G_2$ and H, respectively are

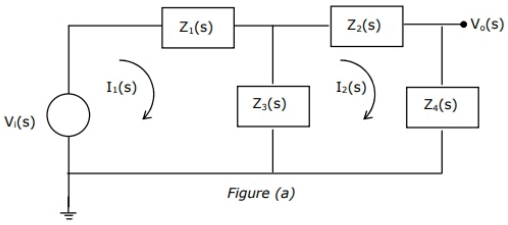
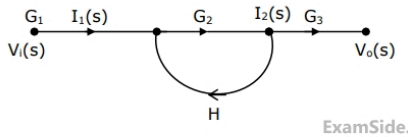



Figure (a)




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Figure (b)

(A) $\frac{Z_3(s)}{Z_2(s)+Z_3(s)+Z_4(s)}, \frac{-Z_3(s)}{Z_1(s)+Z_3(s)}$

(B) $\frac{-Z_3(s)}{Z_2(s)-Z_3(s)+Z_4(s)}, \frac{-Z_3(s)}{Z_1(s)+Z_3(s)}$

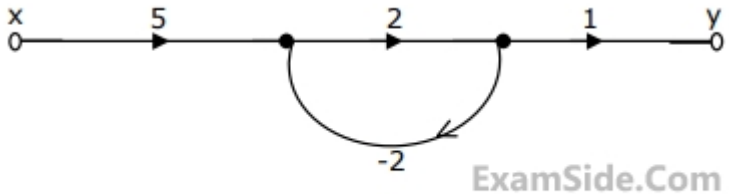
(C) $\frac{Z_3(s)}{Z_2(s)+Z_3(s)+Z_4(s)}, \frac{Z_3(s)}{Z_1(s)+Z_3(s)}$

(D) $\frac{-Z_3(s)}{Z_2(s)-Z_3(s)+Z_4(s)}, \frac{Z_3(s)}{Z_1(s)+Z_3(s)}$

**Q17.** GATE ECE 1997

Open solution

In the signal flow graph of Fig. y/x equals




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(A) 3

(B) 5/2

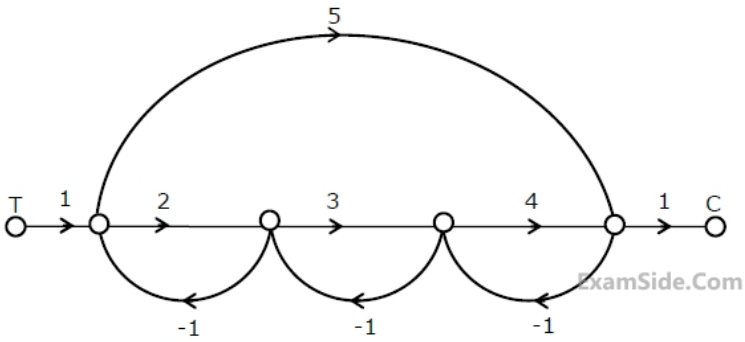
(C) 2

(D) None of the above

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**Q18.** GATE ECE 1991

In the signal flow graph of Figure, the gain c/r will be

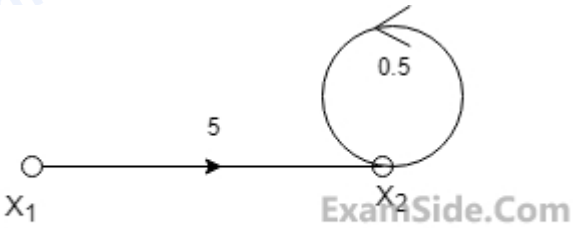



Open solution

(A) 11/9 (B) 22/15 (C) 24/23 (D) 44/23

**Q19.** GATE ECE 1987

In the signal flow graph shown in fig $X_2 = TX_1$ where T is equal to




Open solution

(A) 2.5 (B) 5 (C) 5.5 (D) 10

**Q20.** GATE ECE 1987

For the system shown in figure the transfer function $\frac{C(s)}{R(s)}$ is equal to

GATE ECE 1987 Control Systems - Signal Flow Graph and Block Diagram Question 10 English

Open solution

(A) $\frac{10}{s^2 + s + 10}$ (B) $\frac{10}{s^2 + 11s + 10}$

(C) $\frac{10}{s^2 + 9s + 10}$ (D) $\frac{10}{s^2 + 2s + 10}$

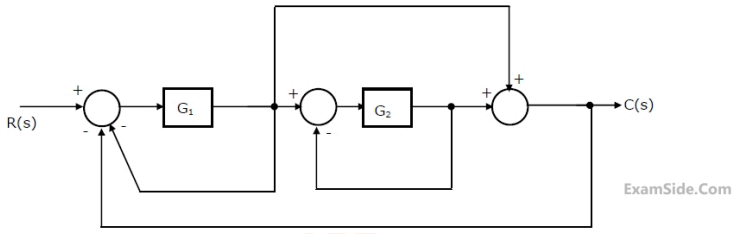
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## Marks 5 3 questions

**Q21.** GATE ECE 2001

Open solution

A feedback control system is shown in figure




(a) Draw the signal-flow graph that represents the system.
(b) Find the total number of loops in the graph and determine the loop-gains of all the loops.
(c) Find the number of all possible combination of non-touching loops taken two at a time.
(d) Determine the transfer function of the system using the signal-flow graph.

**Q22.** GATE ECE 1998

Open solution

Draw a signal flow graph for the following set of algebraic equations:

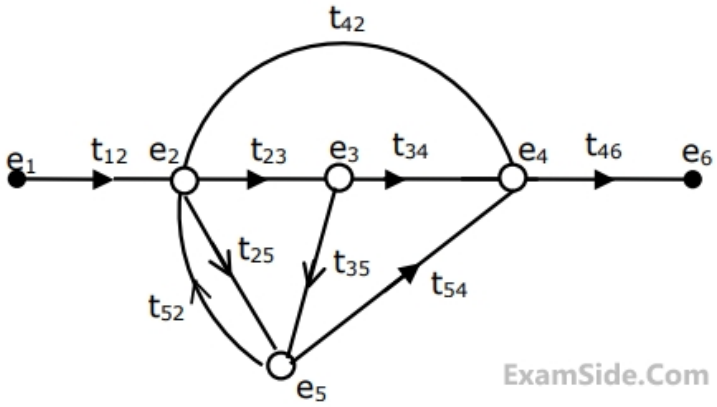
$$y_2 = ay_1 - gy_3$$
$$y_3 = ey_2 + cy_4$$
$$y_4 = by_2 - dy_4$$

Hence, find the gains $\frac{y_2}{y_1}$ and $\frac{y_3}{y_1}$.

**Q23.** GATE ECE 1994

Open solution

Reduce the signal flow graph shown in fig. below, to obtain another graph which does not contain the node $e_5$.Also, remove any self-loop from the resulting graph.




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# Answer Key

## Marks 1

| | | | | |
|---|---|---|---|---|
| 1. **B** | 2. **B** | 3. **B** | 4. **B** | 5. **D** |
| 6. **C** | 7. **B** | 8. **D** | 9. **C** | |

## Marks 2

| | | | | |
|---|---|---|---|---|
| 10. **B** | 11. **A** | 12. **B** | 13. **A** | 14. **C** |
| 15. **D** | 16. **C** | 17. **C** | 18. **D** | 19. **D** |
| 20. **B** | | | | |

## Marks 5

| | | |
|---|---|---|
| 21. **Solve by yourself** | 22. $\frac{y_2}{y_1} = \frac{a(1+d)}{1+d+eg+bcg+edg}$<br>$\frac{y_3}{y_1} = \frac{ae(1+d)+abc}{1+d+eg+bcg+edg}$ | 23. **Try yourself.** |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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- Timed sections, just like exam day
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ANALYSIS

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